

4(a)	any three from: 1 blood arriving / blood from pulmonary artery, has low partial pressure of oxygen / AW ; but deoxygenated blood <i>must be clear that it is blood arriving</i> 2 (so) steep diffusion gradient (for oxygen) ; A high rate of diffusion <i>if mp1 and mp2 not gained, allow 1 mark for correct explanation referencing, diffusion / diffusion gradient / concentration gradient, for oxygen</i> at 13.2 to 13.3 kPa partial pressure in alveolar air 3 haemoglobin has, high / higher, <u>affinity</u> for oxygen ; <i>be flexible with 'for' 'of'</i> 4 allows, percentage saturation of haemoglobin with oxygen to be 98% / haemoglobin (molecule) to bind (all) 4, oxygen molecules / O₂; <i>context is almost all molecules each bind 4 O₂</i> R 4 oxygen atoms A 8 oxygen atoms A each haem group binds an oxygen molecule <i>allow abbreviation for haemoglobin as it is stated in the question text</i> 5 <i>idea that</i> formation of oxyhaemoglobin / oxygen binding to haemoglobin, (removes oxygen) to maintain diffusion gradient ; 6 AVP ; <i>ref. to</i> positive cooperativity / allosteric effect / described / binding of O₂ makes it easier for 2nd to bind owing to conformational change	3
4(b)(i)	(2.4 kPa = 31 / 32 5.6 kPa = 76 / 77) correct values for percentage saturation extracted ; correct answer ; <i>decreases by</i> 76 – 31 or 77 – 32 = 45 (%) or 77 – 31 = 46 (%) or 76 – 32 = 44 (%)	2
4(b)(ii)	any one relevant e.g.: metabolic rate / respiration rate / activity, of, tissue / cells ; rate of blood flow / AW (in capillary network) ; AVP ;	1

Question	Answer	Marks
4(c)	vena cava ; I superior / inferior A venae cavae R vana cava	1

Question	Answer	Marks
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